

1. Data & Labels

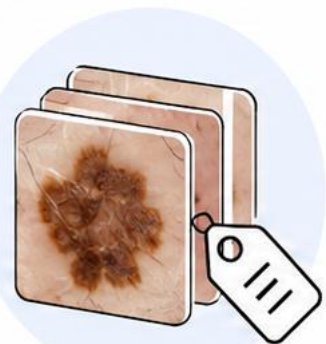
2. Preprocessing & QA

3. Stratified Group K-Fold

4. Training & Early Stopping

5. Evaluation

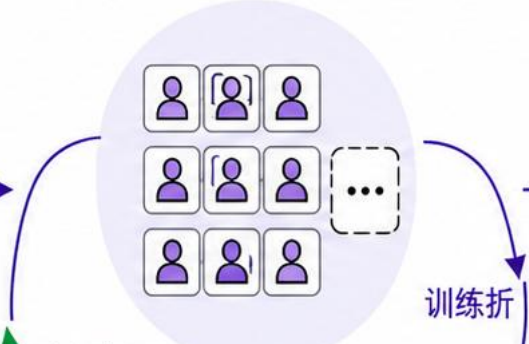
6. Interpretation



原始图像
元数据 / 标签

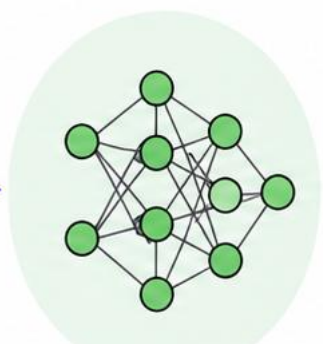


manifest 合并
质量校验

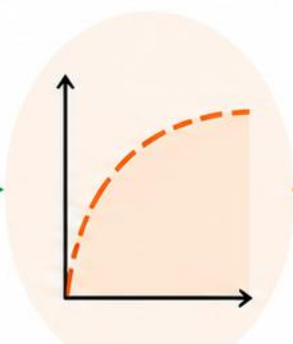


验证折

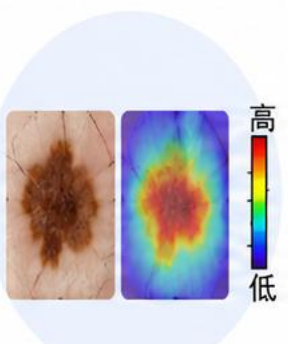
训练折



M0-M6
训练与早停



pAUC / AUPRC
校准 / 公平性



Streamlit / Gradio
可解释展示



病灶异构图

属性 / 患者 / kNN 边

图像 / 表格

预处理